SOFTWARE REQUIREMENTS SPECIFICATION

CareConnect Patient-Provider EHR

Electronic Health Record, e-Prescribing and Patient Portal

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Revision History

Version	Date	Description
0.1	Sep 2026	Initial draft: scope, user roles and module list
0.5	Sep 2026	Functional requirements for all modules, data model
1.0	Oct 2026	Baseline aligned with the implemented and tested system

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1. Introduction

1.1 Purpose

This Software Requirements Specification describes the functional and non-functional requirements of **CareConnect**, a web-based Electronic Health Record (EHR) system that connects patients with their care providers. It is written for the project guide and evaluators, the developers who build and maintain the system, and the testers who verify it. The document is the reference for design, implementation, testing and acceptance of the system.

1.2 Scope

CareConnect supports the day-to-day work of an outpatient clinic. It lets receptionists register patients and book appointments, doctors document visits and place orders, nurses record vitals and administer medicines, pharmacists verify and dispense prescriptions, administrators manage users and master data, and patients view their own records online.

In scope: patient registration and history, clinical notes (SOAP), vital signs, ICD-10 diagnoses, computerized provider order entry (CPOE) with clinical decision support (CDS), laboratory results, e-prescribing, pharmacy verification and dispensing, drug inventory, medication administration record (MAR), appointment scheduling, patient portal with PDF downloads, secure messaging, notifications, dashboards and audit logging.

Out of scope: billing and insurance claims, inpatient bed management, integration with external laboratory or pharmacy systems (HL7/FHIR), real video calls for telehealth, and SMS/e-mail delivery.

1.3 Definitions and Abbreviations

Term	Meaning
EHR	Electronic Health Record – digital version of a patient's medical chart
MRN	Medical Record Number – unique patient identifier, format `CC-YYYY-NNNNNN`
SOAP	Subjective, Objective, Assessment, Plan – standard structure of a clinical note
CPOE	Computerized Provider Order Entry – electronic ordering of tests, medicines and referrals
CDS	Clinical Decision Support – automatic safety checks (allergy, interaction, duplicate)
MAR	Medication Administration Record – log of each dose given to a patient
ICD-10	International Classification of Diseases, 10th revision – diagnosis coding system
e-Rx	Electronic prescription
RBAC	Role-Based Access Control
JWT	JSON Web Token – signed token used to authenticate API requests
Encounter	One visit of a patient with a doctor, containing notes, diagnoses, vitals and orders

1.4 References

- IEEE Std 830-1998, *Recommended Practice for Software Requirements Specifications*
- WHO, *ICD-10 International Statistical Classification of Diseases*
- Project documents: `docs/architecture.md` (diagrams), `docs/api.md` (REST API), `docs/test-cases.md`

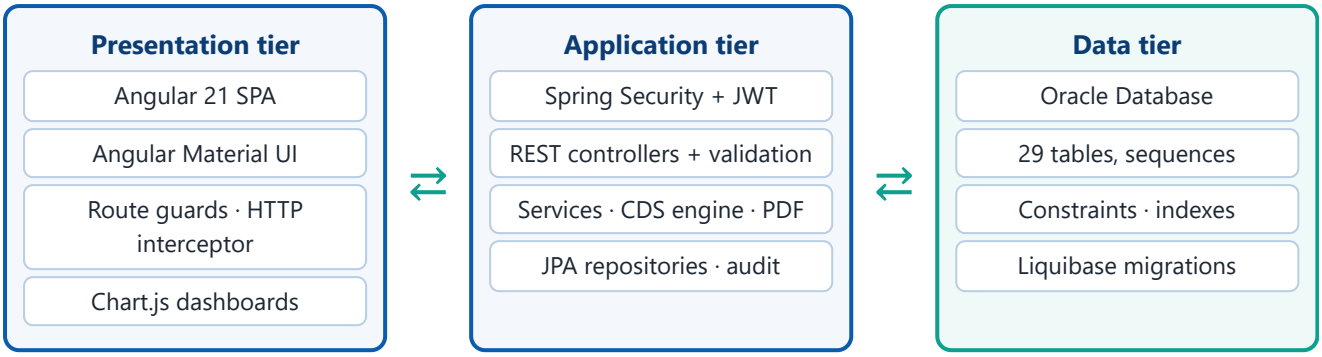
1.5 Document Overview

Section 2 gives an overall description of the product and its users. Section 3 lists the functional requirements module by module, each with a unique identifier and priority. Sections 4–6 cover interfaces, quality attributes and data. Section 7 describes the key use cases and Section 8 the acceptance criteria.

2. Overall Description

2.1 Product Perspective

CareConnect is a new, self-contained system built as a three-tier web application. A single-page Angular application runs in the browser and talks to a Spring Boot REST API over HTTPS using JSON. The API stores all data in an Oracle database and uploaded documents on the server file system.



2.2 Product Functions

- Register patients, generate MRNs and maintain allergies, problems, immunizations, family history and documents.
- Record encounters with SOAP notes, e-signature, addenda, ICD-10 diagnoses and vital signs with trend charts.
- Place lab, imaging, medication and referral orders with automatic allergy, interaction and duplicate checks.
- Enter and flag laboratory results against normal and critical ranges.
- Route e-prescriptions to the pharmacist for verification and dispensing with automatic stock deduction.
- Record every medication dose on the MAR.
- Manage provider slots and appointments, including online booking by patients.
- Give patients a portal with their medicines, results, visit summaries (PDF), bookings and messages.
- Provide role dashboards, in-app notifications, user and master-data administration, and a complete audit trail.

2.3 User Classes and Characteristics

User class	Description and main tasks	Technical skill
Doctor	Opens encounters, writes and signs notes, adds diagnoses, places orders and e-prescriptions, overrides CDS alerts with a reason, reviews results, messages patients.	Moderate
Nurse	Records vital signs, allergies and immunizations, enters lab results, administers medicines and documents them on the MAR.	Moderate
Pharmacist	Verifies or rejects prescriptions, dispenses medicines, maintains the drug formulary, stock and interaction table.	Moderate
Receptionist	Registers patients, edits demographics, manages slots, books, checks in and cancels appointments.	Basic
Administrator	Manages user accounts, roles, departments, ICD-10 codes and lab tests; reviews audit logs.	High
Patient	Views own record, downloads PDFs, books and cancels appointments, messages doctors.	Basic

2.4 Operating Environment

- **Client:** any modern browser (Chrome, Edge, Firefox, Safari) on desktop, tablet or phone.
- **Server:** Java 17 or later; Windows or Linux; minimum 2 GB RAM.
- **Database:** Oracle Database (tested with Oracle XE; later versions supported) on port 1521.
- **Build tools:** Maven (wrapper included) and Node.js 20+ with npm.

2.5 Design and Implementation Constraints

- Backend: Java 17+, Spring Boot 3.x (Web, Data JPA, Security, Validation), Maven, Lombok, MapStruct.
- Frontend: Angular 17+ with standalone components, reactive forms, guards, interceptors and Angular Material.
- Database: Oracle only, using sequences for keys, CHECK/unique/foreign-key constraints and indexes; schema changes via Liquibase.
- Security: JWT access and refresh tokens, BCrypt password hashing, role-based access on both UI and API.
- API documented with OpenAPI (Swagger UI); no secrets hard-coded – configuration through environment variables and profiles.

2.6 Assumptions and Dependencies

- An Oracle database and a schema user with create-table/sequence rights are available before installation.
- Drug, interaction, ICD-10 and lab reference data are loaded as master data and maintained by administrators/pharmacists.
- Users access the system over a trusted network or HTTPS in production.
- The system clock of the server is correct (appointments, audit and token expiry depend on it).

3. Functional Requirements

Priority: **High** = essential for release, **Medium** = important, Low = desirable.

3.1 Authentication and Access Control

ID	Requirement	Priority
FR-01	The system shall authenticate users with username and password and issue a short-lived access token (15 min) and a refresh token (7 days).	High
FR-02	The system shall store passwords only as BCrypt hashes and give the same generic error for unknown user and wrong password.	High
FR-03	The system shall rotate the refresh token on each use and revoke all sessions of a user when a used token is presented again.	High
FR-04	The system shall restrict every screen and API operation to the roles allowed for it (Appendix A); disallowed calls return HTTP 403.	High
FR-05	The client shall renew an expired access token automatically and redirect to login when the session cannot be renewed.	Medium
FR-06	Users shall be able to change their own password (8–72 characters with a letter and a digit) and log out.	Medium

3.2 Patient Records

ID	Requirement	Priority
FR-07	Receptionists and doctors shall register patients with name, date of birth, gender, phone, address, emergency contact and insurance; the system shall generate a unique MRN.	High
FR-08	Staff shall search patients by name, MRN, date of birth or phone with paged and sortable results.	High
FR-09	Doctors and nurses shall maintain allergies (allergen, type, reaction, severity, status), immunizations and family history; doctors shall maintain the problem list with ICD-10 codes.	High
FR-10	Authorised staff shall upload clinical documents (PDF, images, up to 10 MB) with a category, and download them.	Medium
FR-11	The patient chart shall show a banner with identity and active allergies, and a summary of active problems, current medicines, latest vitals and open encounters.	High
FR-12	Updates shall be rejected when the record was changed by another user since it was loaded (optimistic locking).	Medium
FR-13	Administrators shall delete patient records by soft delete only; deleted records are hidden but retained for audit.	Medium

3.3 Clinical Documentation

ID	Requirement	Priority
FR-14	Doctors and nurses shall open an encounter (type, chief complaint, responsible doctor), optionally linked to an appointment.	High

FR-15	Doctors shall write SOAP notes, optionally starting from a template; a note is saved as DRAFT and only its author may edit it.	High
FR-16	Signing a note shall lock it permanently; corrections shall be added as signed addenda linked to the original note.	High
FR-17	Doctors shall add ICD-10 diagnoses to an encounter with type-ahead search; only one diagnosis may be PRIMARY.	High
FR-18	Doctors and nurses shall record vital signs (BP, pulse, temperature, respiratory rate, SpO_2, weight, height); the system shall reject implausible values and calculate BMI.	High
FR-19	The system shall display vital sign trends as a chart and table.	Medium
FR-20	An encounter can be closed only when all its notes are signed; closed encounters accept no new notes or orders.	High

3.4 Order Entry (CPOE) and Clinical Decision Support

ID	Requirement	Priority
FR-21	Doctors shall place lab, imaging, medication and referral orders on an open encounter with priority (routine, urgent, STAT); each order receives a number `ORD-NNNNNN`.	High
FR-22	Before a medication order is placed, the system shall check it against the patient's active drug allergies (by drug name, generic name and class) and raise a CRITICAL alert on a match.	High
FR-23	The system shall check new medicines against the patient's active medicines using the interaction table: MINOR = info, MODERATE = warning, MAJOR = critical, CONTRAINDICATED = hard stop.	High
FR-24	The system shall warn about duplicate orders (same active drug or same pending test).	High
FR-25	Warning and critical alerts shall require an override reason; contraindicated combinations cannot be overridden. Alerts and reasons shall be stored with the order.	High
FR-26	Orders shall follow the lifecycle DRAFT → PLACED → IN_PROGRESS → COMPLETED, or CANCELLED with a reason; backward moves are not allowed.	High
FR-27	Doctors and nurses shall enter results; numeric lab values shall be flagged NORMAL, LOW, HIGH, CRITICAL_LOW or CRITICAL_HIGH against the test's ranges, and the ordering doctor notified.	High

3.5 Medication Management

ID	Requirement	Priority
FR-28	A medication order shall create an e-prescription (`RX-NNNNNN`) with dose, route, frequency, duration and quantity in status PENDING_VERIFICATION, and notify pharmacists.	High
FR-29	Pharmacists shall see the prescription with its CDS alerts, override reason, patient allergies and current stock, and verify it or reject it with a reason.	High
FR-30	Only verified prescriptions can be dispensed; dispensing shall reduce stock, write an inventory ledger entry and complete the order. Dispensing more than the available stock shall be refused.	High
FR-31	When stock reaches the reorder level the system shall list the drug as low stock and notify pharmacists.	Medium

FR-32	Pharmacists and administrators shall maintain the drug formulary, receive or adjust stock, and maintain drug-drug interactions.	Medium
FR-33	Nurses shall document each dose on the MAR as GIVEN, HELD, REFUSED or MISSED; a reason is mandatory for all except GIVEN.	High
FR-34	Doctors shall discontinue a medicine with a reason, or modify it (the old prescription is discontinued and a replacement goes back for verification).	Medium

3.6 Scheduling

ID	Requirement	Priority
FR-35	Authorised staff shall generate equal-length appointment slots for a doctor over a date range (Sundays skipped) and block or unblock slots.	Medium
FR-36	Staff shall book a free slot for a patient; the same slot shall never be booked twice and a patient shall not hold overlapping bookings.	High
FR-37	Staff shall check in, complete, mark no-show or cancel (with reason) an appointment; a checked-in appointment can start an encounter.	Medium

3.7 Patient Portal

ID	Requirement	Priority
FR-38	Patients shall view only their own profile, allergies, conditions, medicines, lab results, vitals and visit summaries.	High
FR-39	Patients shall download a visit summary and a prescription as PDF documents.	Medium
FR-40	Patients shall book free slots with a doctor (in person or telehealth) and cancel their own bookings with a reason.	High

3.8 Messaging, Notifications and Dashboards

ID	Requirement	Priority
FR-41	Patients and doctors shall exchange secure messages grouped by conversation, with unread counts and read status.	Medium
FR-42	The system shall create in-app notifications for new prescriptions, rejections, results, low stock, messages and bookings, shown in a bell menu with unread count.	Medium
FR-43	Each role shall have a dashboard with statistic cards, charts and a work list relevant to that role.	Medium

3.9 Administration and Audit

ID	Requirement	Priority
FR-44	Administrators shall create, edit, enable/disable user accounts, assign roles and departments, reset passwords, and link a patient account to exactly one patient record.	High
FR-45	Administrators shall maintain departments, ICD-10 codes and lab tests (units, normal and critical ranges).	Medium
FR-46	The system shall log every access to patient data and every login/failed login with user, role, action, patient, path, status, IP and time; administrators shall search this log.	High

4. External Interface Requirements

4.1 User Interfaces

- Public landing page describing the product, and a login page with show/hide password.
- After login: side navigation filtered by role, top bar with role badge, notification bell and user menu.
- Consistent Material Design components; tables with paging; dialogs for data entry; colour-coded status chips (e.g. red for critical).
- Confirmation dialogs before irreversible or safety-relevant actions (sign, verify, dispense, cancel, delete, override).
- Validation messages next to each form field; responsive layout down to 360 px width.

4.2 Software Interfaces

- REST API over HTTP(S) with JSON bodies, documented with OpenAPI 3 (Swagger UI at `/swagger-ui.html`); 122 endpoints.
- Oracle Database through JDBC (Oracle thin driver) and JPA/Hibernate.
- Uniform error response: status, error code, message, field errors and details (e.g. CDS alerts).

4.3 Communication Interfaces

- HTTPS in production; header `Authorization: Bearer <token>` on every protected request.
- Cross-origin requests allowed only from configured front-end origins (CORS).

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Security	All API endpoints except login and token refresh require a valid JWT; role checks are enforced on the server with `@PreAuthorize`, and mirrored by route guards in the UI.
NFR-02	Security	Passwords are hashed with BCrypt; refresh tokens are stored only as SHA-256 hashes; secrets (DB password, JWT key) come from environment variables, never from source code.
NFR-03	Privacy	"Minimum necessary" access: e.g. receptionists see demographics but not clinical data; patients can reach only their own record (other records return 404).
NFR-04	Auditability	Audit entries are written in a separate transaction so they are kept even when the business operation fails.
NFR-05	Performance	Typical API responses complete within 1 second for up to 50 concurrent users; list screens are paginated (default 20 rows) and search columns are indexed.
NFR-06	Reliability	Each business operation runs in a database transaction; optimistic locking prevents lost updates; database constraints guarantee that stock never becomes negative.
NFR-07	Data retention	Clinical records are never physically deleted by users (soft delete); signed notes are immutable.
NFR-08	Usability	A new user can complete a core task (e.g. book an appointment) without training; all actions give success/error feedback; forms validate inputs before submission.
NFR-09	Portability	Runs on Windows and Linux; front end works in current versions of major browsers and on mobile screens.

NFR-10	Maintainability	Layered, modular code (controller / service / repository / DTO / mapper per module); database changes only through versioned Liquibase scripts; separate dev, test and prod profiles.
NFR-11	Testability	Automated unit and integration tests (JUnit 5, Mockito, MockMvc) run without an Oracle installation using an in-memory database in Oracle mode.
NFR-12	Availability	The system is stateless on the server side, so it can be restarted or scaled without losing user sessions.

6. Data Requirements

All data is stored in the Oracle schema `CARECONNECT`. Primary keys are generated from one sequence per table; business numbers use dedicated sequences (`mrn_seq`, `order_no_seq`, `rx_no_seq`). Every transactional table carries `created_at/by`, `updated_at/by` and a `version` column.

Group	Tables
Users and security	departments, app_users, refresh_tokens, audit_logs
Patient record	patients, patient_allergies, patient_problems, immunizations, family_history, patient_documents
Clinical	encounters, clinical_notes, encounter_diagnoses, vitals, icd10_codes, note_templates
Orders and results	clinical_orders, order_alerts, lab_tests, lab_results
Medication	drugs, drug_interactions, prescriptions, medication_administrations, inventory_transactions
Scheduling and communication	provider_slots, appointments, messages, notifications

Key integrity rules: unique MRN, username, order number and prescription number; CHECK constraints on all status and enumeration columns (gender, role, order status, severity, etc.); `stock_qty >= 0`; one lab result per order; one prescription per medication order; one portal account per patient; foreign keys between all related tables and indexes on every foreign key and search column.

Master data delivered with the system: 10 departments, about 40 ICD-10 codes, 16 lab tests with normal and critical ranges, 26 drugs, 17 drug interactions and 4 SOAP note templates.

7. Use Case Descriptions

UC-01: Place a medication order with decision support

Actor	Doctor
Precondition	Doctor is logged in; the patient has an open encounter.
Main flow	1. Doctor opens the encounter and selects *New order → Medication*. 2. Doctor searches and selects a drug; the system runs CDS checks immediately. 3. Doctor enters dose, route, frequency, duration and quantity. 4. If there are no alerts, doctor clicks *Place order*; the system saves the order and the e-prescription and notifies pharmacists.
Alternate flow	3a. An allergy, interaction or duplicate alert is shown. The doctor either cancels, or enters an override reason and places the order; the alert and reason are stored. 3b. The combination is contraindicated: the order is blocked and cannot be placed.
Postcondition	Order is PLACED; prescription is PENDING_VERIFICATION.

UC-02: Verify and dispense a prescription

Actor	Pharmacist
Precondition	A prescription is pending verification.
Main flow	1. Pharmacist opens the verification queue and selects a prescription. 2. System shows details, CDS alerts with override reason, patient allergies and stock. 3. Pharmacist verifies (optional note) and confirms. 4. From the dispense queue, pharmacist confirms the quantity; the system deducts stock, writes the ledger and completes the order.
Alternate flow	3a. Pharmacist rejects with a reason; the prescriber is notified. 4a. Stock is insufficient: dispensing is refused with an error message. 4b. Stock falls to reorder level: pharmacists receive a low-stock notification.

UC-03: Document a visit

Actors	Nurse, Doctor
Main flow	1. Receptionist checks in the patient; doctor starts the visit from the appointment. 2. Nurse records vitals; BMI is calculated. 3. Doctor writes a SOAP note from a template and adds ICD-10 diagnoses. 4. Doctor signs the note (confirmation) and closes the encounter.
Exception	4a. An unsigned note exists: closing is refused until all notes are signed. Later corrections are made through signed addenda.

UC-04: Patient books an appointment online

Actor	Patient
Main flow	1. Patient opens *My appointments*, selects a doctor and date. 2. System shows free future slots. 3. Patient picks a time, chooses in-person or telehealth, enters a reason and confirms. 4. System books the slot (via PORTAL) and shows it under upcoming appointments.

Exception	3a. Another user booked the slot meanwhile: the system reports that the slot is taken. Patient may cancel own booking later with a reason.

UC-05: Login and session renewal

Actor	Any user
Main flow	1. User enters username and password. 2. System validates credentials, records the login and returns tokens. 3. User is taken to the role dashboard. 4. When the access token expires, the client obtains a new pair with the refresh token and repeats the request transparently.
Exception	2a. Wrong credentials: generic error, failed login audited. 4a. Refresh token expired or revoked: user is returned to the login page.

8. Acceptance Criteria

The system is accepted when all **High** priority requirements are demonstrated and the following checks pass:

ID	Acceptance test	Covers
`AT-01`	Each of the six roles can log in and sees only its permitted menus; a forbidden API call returns 403.	FR-01–04
`AT-02`	Registering a patient produces an MRN in the format CC-YYYY-NNNNNN; invalid input shows field errors.	FR-07
`AT-03`	A signed note cannot be edited; an addendum can be added; an encounter with a draft note cannot be closed.	FR-15–20
`AT-04`	Ordering Amoxicillin for a Penicillin-allergic patient shows a critical allergy alert and requires an override reason.	FR-22, 25
`AT-05`	Ordering Aspirin for a patient on Warfarin shows an interaction alert; a contraindicated pair is blocked.	FR-23, 25
`AT-06`	A WBC value above the normal range is flagged HIGH and the order becomes COMPLETED.	FR-27
`AT-07`	A prescription cannot be dispensed before verification; dispensing reduces stock by the dispensed quantity.	FR-29–30
`AT-08`	Booking an already booked slot is refused.	FR-36
`AT-09`	A patient can download own visit PDF but gets "not found" for another patient's visit.	FR-38–39
`AT-10`	Opening a patient chart creates an audit entry visible to the administrator.	FR-46

> Verification status of this baseline: 39 automated backend tests, a 76-check API run against Oracle and a browser run of all screens for all roles passed. Detailed test cases are listed in `docs/test-cases.md`.

Appendix A – Role Access Matrix

Function	Admin	Doctor	Nurse	Pharm.	Recep.	Patient
Search patients	✓	✓	✓	✓	✓	–

Register / edit demographics	–	✓	–	–	✓	–
View clinical chart	–	✓	✓	✓	–	own (portal)
Allergies, vitals, immunizations	–	✓	✓	read	–	read own
SOAP notes, diagnoses, close encounter	–	✓	–	read	–	summary
Place / cancel orders	–	✓	–	–	–	–
Enter results	–	✓	✓	–	–	–
Verify / dispense prescriptions	–	–	–	✓	–	–
MAR documentation	–	read	✓	read	–	–
Drug inventory and interactions	✓	read	read	✓	–	–
Appointments and slots	✓	✓	✓	–	✓	own
Messaging	–	✓	–	–	–	✓
Users, departments, master data	✓	–	–	–	–	–
Audit log	✓	–	–	–	–	–

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